FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – III
PAPER –2
TEST DATE: 08-02-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

  Section – A *(01 – 04, 19 – 22, 37 – 40)*: This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

  Section – A *(05 –10, 23 – 28, 41 – 46)*: This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

  Section – B *(11 – 18, 29 – 36, 47 – 54)*: This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	–1	In all other cases.

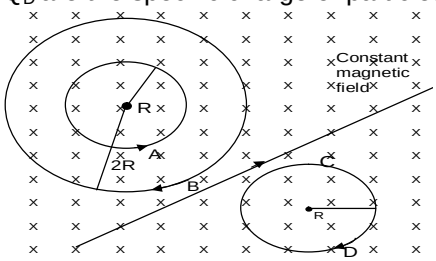
Physics

PART – I

Section – A (Maximum Marks: 12)

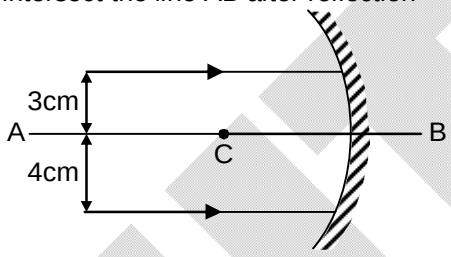
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. Four particles A, B, C and D of masses m_A, m_B, m_C and m_D respectively, follow the paths shown in the figure, in a uniform magnetic field. Each particle moving with same speed. Q_A, Q_B,Q_C and Q_D are the specific charge of particles A,B,C and D respectively :



(A) $Q_A < Q_B < Q_C < Q_D$
(B) $Q_D < Q_B < Q_C < Q_A$
(C) Charge on the particle B and particle D is of different nature
(D) Work done by magnetic force on the particle C is minimum as compared to other particle

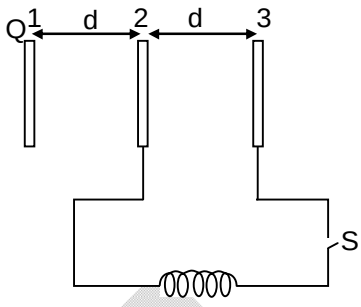
2. Two rays are incident on a spherical concave mirror of radius $R = 5$ cm and rays are parallel to a line AB which passes through the centre C. The rays are at perpendicular distances 3 cm and 4 cm from AB respectively as shown. Find the distance between the points at which these rays intersect the line AB after reflection –



(A) $\frac{25}{24}$cm
(B) $\frac{24}{25}$cm
(C) zero
(D) $\frac{1}{3}$cm

3. A neutron collides head-on with a stationary hydrogen atom $\left({}_1H^1\right)$ in ground state, then choose the correct statement: (Assume that mass of neutron and mass of ${}_1H^1$ atom is same)
(A) If kinetic energy of the neutron is less than 13.6 eV, collision must be elastic
(B) If kinetic energy of the neutron is less than 13.6 eV, collision may be inelastic
(C) Inelastic collision may take place only when initial kinetic energy of neutron is less than 13.6 eV
(D) None of these

4. Three identical large plates are fixed at separation of d from each other as shown. The area of each plate is A. Plate 1 is given charge +Q while plates 2 and 3 are neutral and are connected to each other through coil of inductances L and switch S. If resistance of all connected wires is neglected the maximum current flow through coil after closing the switch is $\left(C = \varepsilon_0 A / d\right)$ (neglect fringe effect)



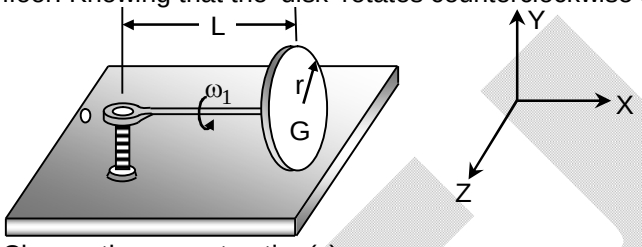
(A) $\frac{Q_0}{\sqrt{LC}}$ (B) $\frac{Q_0}{\sqrt{2LC}}$

(C) $\frac{2Q_0}{\sqrt{LC}}$ (D) $\frac{Q_0}{2\sqrt{LC}}$

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A homogeneous disk of radius r and mass m is mounted on an axle OG of length L and negligible mass. The axle is pivoted at the fixed point O, and the disk is constrained to roll on a horizontal floor. Knowing that the 'disk' rotates counterclockwise at the rate ω_1 about the axle OG,



Choose the correct option(s)

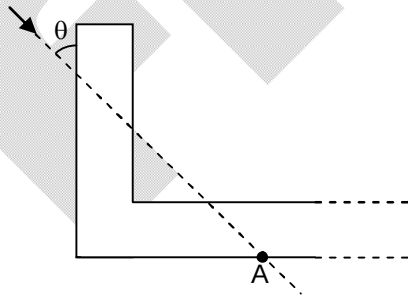
(A) The angular velocity of the disk $\omega_1\hat{i} - \frac{\omega_1 r}{L}\hat{j}$

(B) The angular velocity of the disc $-\left(\frac{\omega_1 r}{L}\right)\hat{j}$

(C) The X component of angular momentum of disc about 'O' is $\frac{1}{2}mr^2\omega_1$

(D) The Z component of angular momentum of disc about 'O' is zero

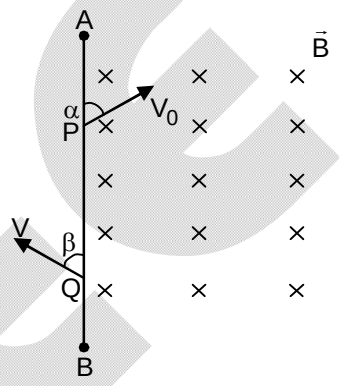
6. Two glass slabs of same thickness and same refractive index are joined to form an 'L' shape as shown. A ray is incident on the combination as shown (in the plane of paper)., Choose correct option(s):



(A) Total deviation will zero for all values of θ.
(B) Total deviation will be zero for $\theta = 45°$ only.
(C) The ray will emerge from point A for all values of θ.
(D) The ray will emerge from point A for $\theta = 45°$ only.

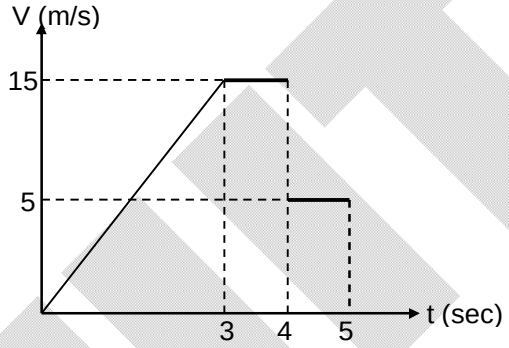
7. If the two slits of double slit experiment were moved symmetrically apart with small relative velocity v and the distance between screen and mid-point of slits is fixed and equal to D. Consider a point P on the screen at a distance x from central maxima then (x<<<D):
(A) Rate of change of number of fringes between central maxima and point P changes with respect to time is $\frac{xv}{\lambda D}$
(B) Number of fringes contained between central maxima and point P increases with time
(C) Fringe width decreases as time passes
(D) Fringe width increases as time passes

8. There exists a uniform magnetic field perpendicular to piane of the paper (inward) right side of the line AB. A particle of charge –q and mass m enters uniform magnetic field B (perpendicular to paper inward) at P with a velocity V_0 at an angle α and leaves the field at Q with velocity V at angle β as shown in figure.



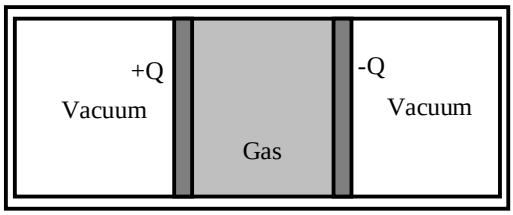
(A) $\alpha = \beta$
(B) $V = V_0$
(C) $PQ = \frac{2mV_0 \sin\alpha}{Bq}$
(D) The particle remains in the field for time $t = \frac{2m(\pi - \alpha)}{Bq}$

9. The figure shows the velocity as a function of the time for an object with mass 10 kg being pushed along a frictionless horizontal surface by external horizontal force. At $t = 3$ s, the force stops pushing and the object moves freely. It then collides head on and sticks to another object of mass 25 kg.



(A) External force acting on the block of mass 10 kg is 50 N from $t = 0$ to $t = 3$ sec.
(B) Speed of the 2nd particle just before the collision is 1 m/s.
(C) Before collision both bodies are moving in the same direction.
(D) Before collision, bodies are moving in opposite direction.

10. A ideal gas having **f** degrees of freedom is kept inside a thermally insulated vessel, the vessel has two large pistons each with charge of +Q and -Q. Assume charges are uniformly distributed. Pistons can move without friction inside the vessel. At an instant the charge of pistons is increased **k** times with the help of an external agent. Then which of the following statements are correct



(A) The Ratio of final to initial temperature of gas in equilibrium is $\frac{K^2+f}{2+f}$.

(B) The Ratio of final to initial temperature of gas in equilibrium is $\frac{2K^2+f}{2+f}$.

(C) Work done by the agent in the process is $(K^2-1)nRT_1$ where T_1 is initial temperature.

(D) Work done by the agent in the process is $\left(K^2-1\right)\frac{Q^2L_1}{A\varepsilon_0}$ where L_1 is initial separation between pistons.

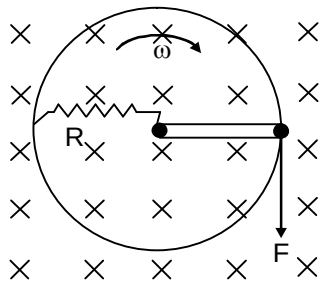
Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

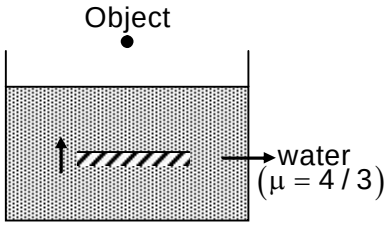
11. A thin spherical soap bubble has surface tension S. Its surface is uniformly charged with charge Q, its volume is V. It is found that when $Q^2=16n\pi\varepsilon_0 SV$, the excess pressure inside the bubble becomes zero. Find n.

12. A circular disc of diameter d lies horizontally inside a metallic hemispherical bowl of radius a. The disc is just visible to an eye looking over the edge. The bowl is now filled with a liquid of refractive index $\mu=\sqrt{3}$. Now, the whole of the disc is just visible to the eye in the same position, if $\frac{d}{a}$ is equal to:

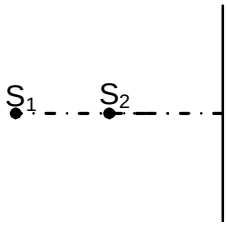
13. A metallic ring of mass 2 kg and radius 1m with a uniform metallic spoke of same mass 2kg and length 1 m is rotated about its axis with angular velocity 1 rev/sec. in a perpendicular uniform magnetic field B of magnitude 10T as shown in figure. If the central end of the spoke is connected to the rim of the wheel through a resistor R magnitude $\pi\Omega$ as shown. The resistor does not rotate, its one end is always at the center of the ring and other end is always in contact with the ring. A force F as shown is needed to maintain constant angular velocity of the spoke then, fill $\frac{F}{25}$ (The ring and the spoke has zero resistance)



14. Mirror in the arrangement shown in figure is moving up with speed 8 cm/sec. Speed of final image (in cm/sec) is 4x, calculate x.



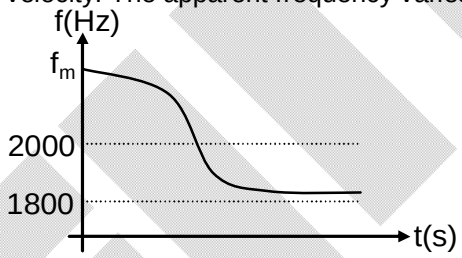
15. Two coherent monochromatic point sources S_1 and S_2 are placed in front of an infinite screen as shown in figure. Wavelength of the light emitted by both the sources is λ. Initial phase difference between the source is zero.



Initially $S_1S_2 = 2.5\lambda$ and the number of bright circular rings on the screen in n_1. If the distance S_1S_2 is increased and made 5.7λ the number of bright circular rings becomes n_2. The difference $n_2 - n_1$ is:

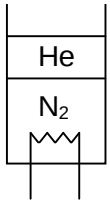
16. If the frequency of k_α x-ray emitted from the element with atomic number 31 is f, if the frequency of k_α x-ray emitted from the element with atomic number 51, is given by $\frac{25}{x}f$, then x is (assume that screening constant for K_α is 1)

17. A stationary observer receives a sound from a source of frequency 2000 Hz moving with constant velocity. The apparent frequency varies with time as shown in the figure.



The value of f_m is (2300 – 10x)Hz. Find the value of x. (Take speed of sound = 300 m/s and neglect the time taken by sound to reach the stationary observer).

18. 5 moles of Nitrogen gas are enclosed in an adiabatic cylindrical vessel. The piston itself is a rigid light cylindrical container containing 3 moles of Helium gas. There is a heater which gives out a power 100 cal/second to the nitrogen gas. A power of 30 cal/sec is transferred to Helium through the bottom surface of the piston. The rate of increment of temperature of the nitrogen gas (in kelvin/sec) is (assuming that the piston moves slowly):



Chemistry

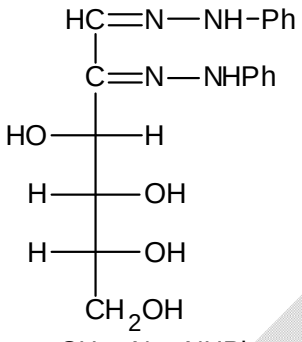
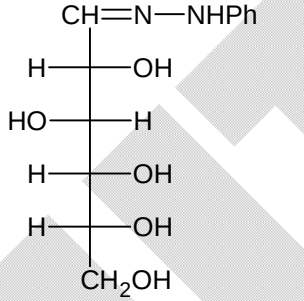
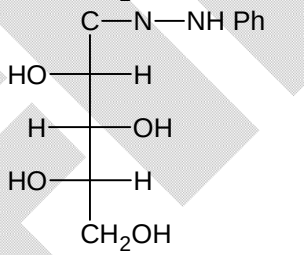
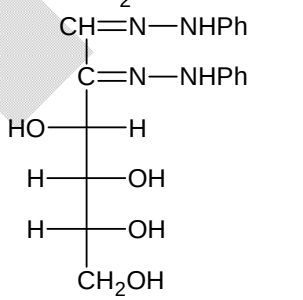
PART – II

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

19. The magnetic moment of $\left[Mn(CN)_6\right]^{3-}$ is 2.8 BM while the magnetic moment of $\left[MnBr_4\right]^{2-}$ is 5.9 BM. The geometries of the complex ions respectively are:
(A) Tetrahedral, tetrahedral
(B) Octahedral, octahedral
(C) Octahedral, tetrahedral
(D) Tetrahedral, octahedral

20. D-glucose on treatment with x mol of $PhNHNH_2$ gives 1 mol of (A), 1 mol of $PhNH_2$ and 1 mol of NH_3. (A) and x are:

	A	x
(A)		3
(B)		3
(C)		3
(D)		2

21. Initially 3 moles of 'A' was taken in a 1L container. What would be the approx. moles of A left in the container when the following equilibrium is established.

$3A \rightleftharpoons B; \quad K_C = 8 \times 10^{15}$

(A) 1.67×10^{-4} (B) 3.33×10^{-8}

(C) 1.25×10^{-10} (D) 5×10^{-6}

22. Which of the following cationic detergents?

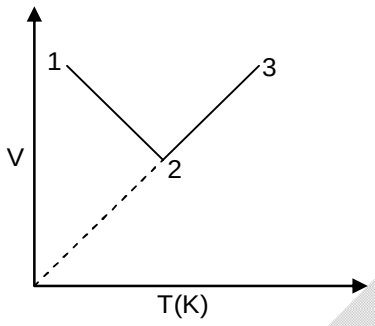
(A) Sodium lauryl sulphate (B) Cetyl trimethyl ammonium bromide.

(C) Sodium dodecylbenzene sulphonate (D) Glyceryl oleate

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

23. Following graph is constructed for the fixed amount of the gas. Which of the following is/are correct?



(A) From $1 \rightarrow 2$ pressure will increase

(B) From $2 \rightarrow 3$ pressure remains constant

(C) Gas pressure at (3) is greater at state (1)

(D) For pressure the correct order is $P(3) > P(2) > P(1)$

24. The compound(s) which (have) peroxo linkage is (are):

(A) $H_2S_2O_3$ (B) H_2SO_5

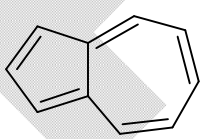
(C) $H_2S_2O_7$ (D) $H_2S_2O_8$

25. Which one is classified as addition polymer?

(A) Acrylonitrile (B) Dacron

(C) Neoprene (D) Teflon

26. Which of the following is /are true about azulene?



Azulene

(A) Both rings are aromatic in nature

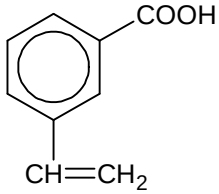
(B) It is soluble in polar solvents

(C) Attack of electrophile occurs at five membered ring

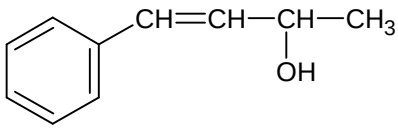
(D) Attack of nucleophile occurs at seven membered ring

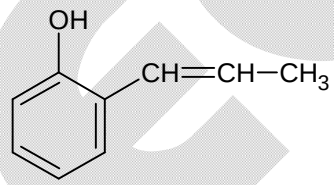
27. Which one of the following statements about acidic/ basic strength is/are correct?
(A) HCOOH is more acidic than $Ph-COOH$
(B) Acidic strength odder is as $CH_3OH > H_2O > C_2H_5-OH$
(C) Fumaric acid is stronger acid than Maleic acid
(D) Pyridine is more basic than pyrrole

28. Compound 'P' $(C_{10}H_{12}O)$ evolves H_2 gas with Na metal. It reacts with Br_2/CCl_4 to give 'Q' $(C_{10}H_{12}Br_2O)$. With I_2/NaOH it forms iodoforms and an acid 'R' $(C_9H_8O_2)$. 'P' has geometrical and optical isomers. The structure of 'P' and 'R' should be

(A) 

(B) $PH-CH=CH-COOH$

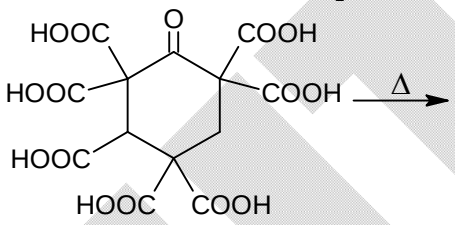
(C) 

(D) 

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. The number of moles of CO_2 that will be evolved in the following is



30. How many of the reagents from the lists below would give effeverence when treated with propyne?

NaOH (I)	CH_3CH_2ONa (II)
CH_3CH_2MgBr (III)	NaH (IV)
$NaNH_2(V)$	Na (VI)
$NaHCO_3(VII)$	$\{(CH_3)CH\}_2NLi(VIII)$
$CH_3CH_2Li(IX)$	

31. How many of the following are reducing sugars?

Glucose	Fructose	Mannose	Sucrose	Maltose	Cellobriose
Lactose	Cellulose	Starch			

32. The formation constant of complex HgI_4^{2-} is $1.0\times10^{(n\times5)}$
Given: $E^\circ_{\left(Hg^{2+}/Hg\right)} = 0.85$ and $E^\circ_{\left(HgI_4^{2-}/Hg,\ I^-\right)} = -0.05$ V.
The value of 'n' is

33. 30 ml of 0.2 M NaOH is added to 50 ml 0.2 M CH_3COOH is give 80 ml of the solution. $\frac{10}{x}$ mL of 0.2 M NaOH required to make the pH of the solution 5.00. The value of 'x' is
Given: The ionization constant of $CH_3COOH = 2\times10^{-5}$.

34. Iron exists as BCC at 10°C while exist as FCC at 910°C. What would be the percentage decrease in volume of iron at 910°C from 10°C?

35. Among the following how many of the molecule is linear?
CO_2, NO_2^-, SO_2, N_3^-, C_2H_2, I_3^-, XeF_2, $BeCl_2$, CS_2

36. For the reaction: $3A_{(g)} \longrightarrow 2B_{(g)}$, the rate of formation of 'B' at 298 K is represented as $\ln\left(\frac{d[B]}{dt}\right) = -0.04 + 2\ln[A]$. The order of reaction is

Mathematics

PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. If $(\alpha-\beta)$, $(\alpha+\beta)$, $\left(\frac{1}{\alpha}+\frac{1}{\beta}\right)$ are in A.P. for $\alpha>0>\beta$ and $|\alpha|>|\beta|$, then for quadratic equation $ax^2+bx-1=0$

(A) $a\in(-2,\infty)$ (B) $a\in(-\infty,2)$

(C) $a\in(-\infty,0)$ (D) $a\in(0,\infty)$

38. $\lim\limits_{n\to\infty}\sum\limits_{r=3}^{n}\frac{(r^3-8)}{(r^3+8)}=$

(A) $\frac{1}{3}$ (B) $\frac{2}{5}$

(C) $\frac{3}{8}$ (D) $\frac{2}{7}$

39. Let P be a point (not origin) lying on the parabola $x^2=y$. The normal line to the parabola at P will intersect the parabola at another point Q. The minimum possible value for area bounded by the line PQ and the parabola is

(A) $\frac{2}{3}$ (B) $\frac{3}{4}$

(C) $\frac{4}{3}$ (D) $\frac{3}{5}$

40. If $\frac{(\tan^2 x)}{\left(1+\tan x-\sqrt{1+\tan x}\right)^2}<\frac{\sec^2 x+3\tan x+17}{(1+\tan x)^3}$ for $-\frac{\pi}{2}<x<\frac{\pi}{2}$, then number of solutions of $\tan x+x^2=1$ is/are

(A) 0 (B) 1

(C) 2 (D) Infinite

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

41. Which is/are correct?

(A) $\int_0^{\pi}\frac{\sin^3 x\,dx}{(1+x)}=3\int_0^{\pi}\sin 2x.(x+1)\left\{\ln\left(\frac{x+1}{e}\right)\right\}dx$ (B) $\int_0^{\pi}\frac{\sin^3 x\,dx}{2x-\pi}=\int_0^{\pi/2}\frac{\cos^3 x}{x}dx$

(C) $\int_0^{\pi}\frac{\sin^3 x\,dx}{(1+x)}+\int_0^{\pi}\frac{\cos^3 x}{x}dx=0$ (D) $\int_0^{\pi}\frac{\sin^3 x\,dx}{(1+x)}=3\int_0^{\pi}\sin 2x(x+1)\ln(x+1)dx$

42. If α, β are roots of $x^2 - x - 1 = 0$, then quadratic whose roots are either $\left(\frac{\alpha^6 - 1}{\alpha^3}\right)$ or $\left(\frac{\beta^6 - \beta^3 - 1}{\beta^3}\right)$ is/are

(A) $x^2 - 5x + 4 = 0$ (B) $x^2 - 7x + 12 = 0$

(C) $x^2 - 5x + 6 = 0$ (D) $x^2 - 9x + 14 = 0$

43. If $f(n) = \sum_{k=1}^{k=n} \int_k^{k+1} \frac{(k+1)dx}{x(x+1)}$ and $g(x) = (x+1)\ln\left(\frac{x+2}{x+1}\right) + f(x)$, then

(A) $g'(0) = 2$ (B) $g''(0) = -1$

(C) $g'(1) = \frac{1}{2}$ (D) $\lim_{n\to\infty} \sum_{r=1}^{r=n} \left(\frac{1}{2.e^{g(r)} - 1 + n}\right) = \log_e 2$

44. If P moves on a circle which touches both the axis and angle bisector of $\angle APB$ for two different positions of P in the same segment are given by $x + y = 3$ and $x - y = 1$, then radius of circle can be _____ (AB is given chord)

(A) 1 (B) 3

(C) 4 (D) 5

45. Let $f : R \to (0, 1)$ be a continuous function, then which of the following pair of vectors are linearly dependent for some $x \in (0, 1)$?

(A) $\vec{a} = (f(x))\hat{i} + 2\hat{j},\ \vec{b} = x^2\hat{i} + 3\hat{j}$ (B) $\vec{a} = (f(x))\hat{i} + 3\hat{j},\ \vec{b} = x^2\hat{i} + 2\hat{j}$

(C) $\vec{a} = \left(\int_0^{1-x} f(t)dt\right)\hat{i} + 3\hat{j},\ \vec{b} = x\hat{i} + 2\hat{j}$ (D) $\vec{a} = \left(\int_0^{1-x} f(t)dt\right)\hat{i} + 2\hat{j},\ \vec{b} = x\hat{i} + 3\hat{j}$

46. Matrices of order 2×2 are formed by using the elements of the set $A = \{-2, -1, 0, 1, 2\}$. Then, probability that matrix is either symmetric or skew-symmetric, is greater than

(A) $\frac{1}{10}$ (B) $\frac{2}{15}$

(C) $\frac{3}{5}$ (D) $\frac{3}{10}$

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. Number of integral values of a for which $\frac{2x^2 + ax + 2}{x^2 + x + 1} = \frac{e^y}{[y+1]}$ is true $\forall x \in R$ and $y \geq 0$ (where [.] denotes G.I.F), is

48. If det (A) = $\begin{vmatrix} \alpha & 0 \\ 2 & \bar{\alpha} \end{vmatrix}$, where α is complex number, $\bar{\alpha}$ is its conjugate and $\det(2A^2 - 2A) = 144$, then tr(A) is either m or n, then (m + n) = _____.

49. If $f(x) = \frac{\alpha x + 1}{x - \beta}$ $\forall x \in R - \{\beta\}$ & $\alpha\beta \neq 1$ & $\alpha \neq 1$ is a self-inverse function such that $\frac{f(4)}{4} = \frac{f(12)}{12} = f\left(\frac{1+\beta}{1-\alpha}\right)$, then "$\beta$" is equal to

50. If $f(x) = x + 1$ & $g^{-1}(x) = x^3 + x + 1$, let $gof(x) = h(x)$, then $\lim_{x \to 1} \frac{h^{-1}(x) - h^{-1}(1)}{x - 1}$ =

51. Let α be a complex number satisfying $i\alpha^3 + \alpha^2 - \alpha + i = 0$ and β be a complex number satisfying $|\beta - 1| = 4$, then maximum value of $|\alpha + \beta|$ is

52. Let A be a square matrix of order 3 such that transpose of inverse of A is A itself, then $|\text{adj}(\text{adj}A)|$ is equal to

53. If $p = \sin^4\alpha + \cos^4\alpha + 1$ and $q = \cos^6\beta + \sin^6\beta + 1$ $(\alpha, \beta \in R)$, then the value of $\left[(p+q)_{max}\right] + \left[(p+q)_{min}\right]$ is (where [.] is G.I.F)

54. Let $n = x_1x_2x_3 \ldots\ldots x_9$ is formed using the digit $1, 2 \ldots 9$. Each digit is being used once only. If n_1 is number of possible n for which $1 < |x_3 - x_4| < 5$ and n_2 is number of possible n for which $|x_3 - x_4| = 1$, then $\left[\frac{n_1}{n_2}\right]$ is equal to (where [.] is G.I.F)